

A NOTE OF TOPOLOGICAL METHODS IN GROUP THEORY

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ABSTRACT. This is a course note of 2026 Spring MATH 569, taught by Daniel Groves at UIC, on the topic of Topological Methods in Group Theory. Major reference textbooks are: [Hat01], [Bro82], [Geo08].

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1. ASPHERICAL SPACES AND SAMPLE EARLY THEOREMS

In this course we will explore the relationship between aspherical spaces and (discrete) groups. A convention: unless otherwise stated, all spaces in this course will be CW complexes.

Definition 1.1 ($K(G, 1)$). Let G be a group. A $K(G, 1)$ is a connected space such that $\pi_1(K(G, 1)) \cong G$ and whose universal covering space $\widehat{K}(G, 1)$ is contractible.

It is also called *the classifying space* or *Eilenberg-MacLane space*.

Definition 1.2 (Aspherical Space). If X is a connected space such that $\widetilde{X}$ is contractible, then we say X is a $K(\pi, 1)$ (i.e. $K(\pi_1(X), 1)$). We call such X *aspherical*.

Theorem 1.3 ([Hat01, Example 1.B.7, Theorem 1.B.8, Proposition 1.B.9]). *For any group G , there exists a (CW) $K(G, 1)$ which is unique up to homotopy equivalence. In fact, for any two groups G, H , there exists a bijection*

$$\mathrm{Hom}(G, H) \leftrightarrow \{\text{based maps } K(G, 1) \rightarrow K(H, 1)\} / \text{based homotopy}$$

Let X be a space and x_0 a point in X . Recall that $\pi_0(X)$ represents the path components of space X . For $n \geq 1$, $\pi_n(X, x_0)$ is the group of based homotopy classes of maps $(\mathbb{S}^n, \star) \rightarrow (X, x_0)$.

For $n \geq 2$, any map $(\mathbb{S}^n, \star) \rightarrow (X, x_0)$ lifts to $(\mathbb{S}^n, \star) \rightarrow (\widetilde{X}, \widetilde{x}_0)$ (Since $\pi_1(\mathbb{S}^n, \star)$ is trivial). If X is aspherical, then $(\mathbb{S}^n, \star) \rightarrow (\widetilde{X}, \widetilde{x}_0)$ is null-homotopic. Hence $\pi_n(X, x_0) = 0$. This proves one direction of Theorem 1.4. The other direction is harder, and we leave it for future discussion.

Theorem 1.4 (Whitehead's Theorem, [Hat01, Theorem 4.5]). *A CW complex X is aspherical if and only if it's connected and for all $n \geq 2$, $\pi_n(X, x_0) = 0$.*

Example 1.5 (Aspherical Spaces). (1) A point.

(2) $\mathbb{S}^1$.

(3) Products of aspherical spaces (corresponding to direct products of groups).

(4) Wedges of aspherical spaces (corresponding to free products of groups).

(5) Graphs (1-dimensional CW complexes).

(6) Closed orientable surfaces with genus ≥ 1 .

(7) Complete Riemannian manifolds of non-positive sectional curvature (Cartan-Hadamard theorem).

(8) Complete locally CAT(0) spaces (A variant of Cartan-Hadamard theorem. See for example [BH99, Theorem II.4.1]).

Example 1.6 (Non-aspherical Spaces). Connect sums of aspherical spaces typically produce non-aspherical spaces.

Question 1.7. • What kind of $K(G, 1)$ spaces does [insert any interesting group] have?

- For [insert any interesting group], which $K(G, 1)$ is “better”?
- What do properties of $K(G, 1)$ say about G ?

Below we list some sample early theorems without proof.

Theorem 1.8. *A group G is finitely generated if and only if G has a $K(G, 1)$ with a finite 1-skeleton.*

Theorem 1.9. *A group G is finitely presented if and only if G has a $K(G, 1)$ with a finite 2-skeleton.*

Theorem 1.10. *If G has non-trivial torsion, then there does not exist any finite-dimensional $K(G, 1)$.*

Another way to phrase Theorem 1.10 is: if G has a finite-dimensional $K(G, 1)$, then it's torsion free.

Observe that those theorems all related to some sort of finiteness of $K(G, 1)$. We introduce some notions of finiteness below.

Definition 1.11. A group G is of *type F_n* if it has a $K(G, 1)$ with finite n -skeleton.

A group G is of *type F_∞* if it's of type F_n for all n .

A group G is of *type F* if it has a finite $K(G, 1)$.

A group G has *finite geometric dimension* if it has a finite-dimensional $K(G, 1)$. The *geometric dimension*, denoted as $\text{gd}(G)$, is the minimal dimension of all of its $K(G, 1)$.

It is sometimes difficult to understand the $K(G, 1)$ of a group. We thus use multiple invariance to help determine those types. If G is a group and X is its $K(G, 1)$, then the algebro-topological invariance of X is invariant of G . In particular, we look at the cohomology of groups (with trivial coefficients):

$$H^*(G, \mathbb{Z}) := H^*(X, \mathbb{Z})$$

2. COHOMOLOGY OF GROUPS

Definition 2.1. We say a space (Y, y_0) is *n -connected* if $\pi_i(Y, y_0) = 0$ for all $0 \leq i \leq n$.

Other than the trivial coefficient $\mathbb{Z}$, we use also group rings to calculate the cohomology of groups to get different properties.

Definition 2.2. Let G be a group. The group ring $\mathbb{Z}G$ is the set of finite $\mathbb{Z}$ -linear combinations of elements of G .

Remark 2.3. We may also use $\mathbb{R}$, fields, or PIDs, etc. instead of $\mathbb{Z}$ to get different group rings.

Example 2.4. Let $G = \mathbb{Z} = \langle t \rangle$. Then $\mathbb{Z}G = \mathbb{Z}[t, t^{-1}]$.

Example 2.5. Let $G = \mathbb{Z}/n\mathbb{Z} = \langle t \rangle$. Then $\mathbb{Z}G = \mathbb{Z}[t]/(t^n - 1)$.

Notice that $g^n = 1$ for some non-trivial element $g \in G$, i.e. G has a non-trivial torsion, then we have:

$$0 = g^n - 1 = (g - 1)(g^{n-1} + \cdots + g + 1)$$

which implies that $g - 1$ is a non-zero zero-divisor. The other direction of this fact is an open question.

Question 2.6 (Kaplansky's Conjecture). If G is torsion-free, then $\mathbb{Z}G$ has no non-zero zero-divisor.

A motivation for Question 2.6 is that we may embed $\mathbb{Z}G$ into its division ring, so we wish to have no non-zero zero-divisors in $\mathbb{Z}G$.

Typically G is not abelian, so $\mathbb{Z}G$ is not a commutative ring. Left and right $\mathbb{Z}G$ -modules are typically different. By default, we use left $\mathbb{Z}G$ -modules.

Remark 2.7. If M is a left $\mathbb{Z}G$ -module, it can be turned into a right $\mathbb{Z}G$ -module by defining a right G -action $mg = g^{-1}m$.

Definition 2.8 (Free resolution). A free resolution of $\mathbb{Z}$ (the trivial $\mathbb{Z}G$ -module) by free $\mathbb{Z}G$ -module is a resolution

$$\cdots \rightarrow F_1 \rightarrow F_0 \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

that is exact, and every F_i is a free $\mathbb{Z}G$ -module (i.e. $(\mathbb{Z}G)^n$ for some n).

Sometimes we also use resolutions by projective $\mathbb{Z}G$ -modules (See Definition 5.5).

Remark 2.9. The free resolution arises naturally from $K(G, 1)$ in the following manner. Say X is a CW complex $K(G, 1)$ (recall Definition 1.1), then any i -cell in $\tilde{X}$ is $g\sigma$ for some $g \in G$ and some i -cell σ in X . So we can view the cellular i -chain $C_i(\tilde{X}; \mathbb{Z})$ as a free $\mathbb{Z}G$ -module whose basis is in bijective with i -cells in X .

Since $\tilde{X}$ is contractible, $H_*(\tilde{X}) = 0$. That is, the complex

$$\cdots \rightarrow C_{n+1}(\tilde{X}) \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} \cdots \xrightarrow{\partial_1} C_0(\tilde{X}) \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

is exact. The boundary maps are $\mathbb{Z}G$ -module homomorphisms. This gives a free resolution of $\mathbb{Z}$ by free $\mathbb{Z}G$ -modules.

Theorem 2.10. *There is a unique resolution of $\mathbb{Z}$ by free (projective) $\mathbb{Z}G$ -module.*

Definition 2.11 (Cohomology of group). Let M be a left $\mathbb{Z}G$ -module. Let $F = \{(F_i, d_i)\}$ be a free resolution of $\mathbb{Z}$. We truncate F to get a (non-exact) complex:

$$\cdots \rightarrow F_i \xrightarrow{d_i} F_{i-1} \xrightarrow{d_{i-1}} \cdots \xrightarrow{d_1} F_0 \rightarrow 0.$$

Apply $\text{Hom}_{\mathbb{Z}G}(-, M)$ to the complex. Denote $F^i = F_i^* = \text{Hom}_{\mathbb{Z}G}(F_i, M)$. We have:

$$\cdots \leftarrow F^i \xleftarrow{d^{i-1}} F^{i-1} \xleftarrow{d^{i-2}} \cdots \xleftarrow{d^0} F^0 \leftarrow 0.$$

The *cohomology of G* with coefficients in M , denoted as $H^*(G; M)$, is the homology of the complex above:

$$H^i(G; M) = \ker d^i / \text{im } d^{i-1}.$$

If we do not truncate the resolution, then we get the exact sequence:

$$\cdots \leftarrow \text{Hom}_{\mathbb{Z}G}(F_1, M) \xleftarrow{d^0=d_1^*} \text{Hom}_{\mathbb{Z}G}(F_0, M) \xleftarrow{\epsilon^*} \text{Hom}_{\mathbb{Z}G}(\mathbb{Z}, M) \leftarrow 0.$$

Denote $H^0(G; M)$ as M^G . Notice that $M^G = \ker d^0 = \text{im } \epsilon^* \cong \text{Hom}_{\mathbb{Z}G}(\mathbb{Z}, M)$. Recall that G acts on $\mathbb{Z}$ trivially, so elements in $\text{Hom}_{\mathbb{Z}G}(\mathbb{Z}, M)$ have to be elements in M that are invariant under G , i.e. $M^G = \{m \in M \mid \forall g \in G, gm = m\}$.

Since free resolutions are chain homotopic, $H^*(G; M)$ is independent of the choices of the free resolution.

Example 2.12 (Cohomology of $\mathbb{Z}/n\mathbb{Z}$). Let $G = \mathbb{Z}/n\mathbb{Z}$, $M = \mathbb{Z}$. Then $\mathbb{Z}G = \mathbb{Z}[t]/(t^n - 1)$. Consider $\mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$, $\ker \epsilon = (t - 1)$. Let $N = t^{n-1} + \cdots + t + 1$. We have the free resolution:

$$\cdots \rightarrow \mathbb{Z}G \xrightarrow{\cdot N} \mathbb{Z}G \xrightarrow{\cdot (t-1)} \mathbb{Z}G \xrightarrow{\cdot N} \mathbb{Z}G \xrightarrow{\cdot (t-1)} \mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0.$$

Apply $\text{Hom}_{\mathbb{Z}G}(-, \mathbb{Z})$ to the resolution. For any $f \in \text{Hom}_{\mathbb{Z}G}(\mathbb{Z}G, \mathbb{Z})$, $f(\sum_i \alpha_i t^i)$ has to be $k \sum_i \alpha_i$ where $k \in \mathbb{Z}$, i.e. $\text{Hom}_{\mathbb{Z}G}(\mathbb{Z}G, \mathbb{Z}) = \mathbb{Z}$. Thus the cochain complex we obtained is:

$$\cdots \xleftarrow{n} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{n} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \leftarrow 0.$$

We conclude that

$$H^i(\mathbb{Z}/n\mathbb{Z}; \mathbb{Z}) = \begin{cases} \mathbb{Z} & \text{if } i = 0, \\ \mathbb{Z}/n\mathbb{Z} & \text{if } i \geq 1 \text{ is odd,} \\ 0 & \text{if } i \geq 1 \text{ is even.} \end{cases}$$

Proposition 2.13. *If X is an n -dimensional $K(G, 1)$, M is any $\mathbb{Z}G$ -module, then $\forall m > n$, $H^m(G; M) = 0$.*

Proof. The free resolution is trivial for $m > n$. □

Corollary 2.14. *There does not exist any finite-dimensional $K(\mathbb{Z}/n\mathbb{Z}, 1)$ for $n \geq 2$.*

We are now ready to prove Theorem 1.10. Recall the statement:

Theorem 1.10. *If G has non-trivial torsion, then there does not exist any finite-dimensional $K(G, 1)$.*

Proof. Let X be a $K(G, 1)$. For any subgroup $H \leq G$, let X^H be the cover of X corresponding to H . Then X^H is a $K(H, 1)$. Observe that X and X^H have the same dimension.

If G has non-trivial torsion, then there exists $\mathbb{Z}/n\mathbb{Z} \leq G$ with $n \geq 2$. By Corollary 2.14, X^H is infinite-dimensional, hence so is X . $\square$

3. COMPACTLY SUPPORTED CELLULAR COHOMOLOGY

In this course we focus on $\mathbb{Z}G$ -module with trivial coefficient. For coefficients other than $\mathbb{Z}$, a key phrase is “local systems of coefficients on X ”.

Theorem 3.1. *If X is a $K(G, 1)$, then for all $i \geq 0$, $H^i(G; \mathbb{Z}) \cong H^i(X; \mathbb{Z})$.*

Proof. Recall that X gives a free $\mathbb{Z}G$ resolution of $\mathbb{Z}$ (See Remark 2.9):

$$\cdots \rightarrow C_1(\tilde{X}) \xrightarrow{d_1} C_0(\tilde{X}) \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0.$$

Apply $\text{Hom}_{\mathbb{Z}G}(\cdot, \mathbb{Z})$ to the resolution and consider $\text{Hom}_{\mathbb{Z}G}(C_i(\tilde{X}), \mathbb{Z})$. Recall that $\mathbb{Z}G$ homomorphism has to be invariant under G , so the map has to assign the same number to each i -cell in the same G -orbit. It is naturally equivalent to assigning numbers to i -cells in X i.e. cellular chains in X . $\square$

Definition 3.2 (Compactly supported homomorphism). Let M be a $\mathbb{Z}G$ -module. Define $\text{Hom}_c(M, \mathbb{Z}) \subseteq \text{Hom}(M, \mathbb{Z})$ to be

$$\{f : M \rightarrow \mathbb{Z} \mid \forall m \in M, \text{ for all but finitely many } g \in G, f(gm) = 0\}.$$

Lemma 3.3 ([Bro82, Lemma VIII.7.4]). *Let Γ be a $\mathbb{Z}G$ -module, then*

$$\text{Hom}_{\mathbb{Z}G}(M, \mathbb{Z}G) \cong \text{Hom}_c(M, \mathbb{Z}).$$

Proof. Take $F \in \text{Hom}_{\mathbb{Z}G}(M, \mathbb{Z}G)$. For $m \in M$, $F(m)$ can be written as $\sum_{g \in G} f_g(m)g$ where $f_g(m) = 0$ for all but finitely many g .

Verify that $F \in \text{Hom}_{\mathbb{Z}G}(M, \mathbb{Z}G)$ if and only if $f_g(m) = f_1(g^{-1}m)$. So $F \mapsto f_1$ gives a map whose inverse is:

$$f \mapsto \{m \mapsto \sum_{g \in G} f(g^{-1}m)g\}.$$

$\square$

Definition 3.4 (Compactly supported cohomology). Let Y be a locally compact CW complex, i.e. for all n -cell σ , there are finitely many $n+1$ cells whose boundary meets σ . Let $C_c^i(Y; \mathbb{Z}) \subseteq C^i(Y; \mathbb{Z})$ be the set of finitely supported cellular cochains with

$$\delta_c^i : C_c^i(Y; \mathbb{Z}) \rightarrow C_c^{i+1}(Y; \mathbb{Z}).$$

The homology of this chain complex, denoted as $H_c^i(Y; \mathbb{Z})$, is the *compactly supported cohomology*.

Remark 3.5. The requirement of locally compactness is to ensure that finitely supported cochains and compactly supported cochains are the same set.

Theorem 3.6. *If X is a finite $K(G, 1)$, then*

$$H^i(G; \mathbb{Z}G) \cong H_c^i(\tilde{X}; \mathbb{Z}).$$

Proof. Recall that by Lemma 3.3,

$$\text{Hom}_{\mathbb{Z}G}(C_i(\tilde{X}), \mathbb{Z}G) \cong \text{Hom}_c(C_i(\tilde{X}), \mathbb{Z}) = C_c^i(\tilde{X}; \mathbb{Z}).$$

So they induce the same cohomology (See Definitions 2.11 and 3.4). $\square$

Remark 3.7. A more natural definition of compactly supported cohomology arises from the direct limit of relative cohomology groups. Observe that a finitely supported i -cochain is 0 outside some compact set K , so it lies in $C^i(Y, Y \setminus K; \mathbb{Z})$. We define:

$$H_c^*(Y; \mathbb{Z}) = \varinjlim_{K \text{ compact}} H^*(Y, Y \setminus K; \mathbb{Z}).$$

We give some examples of calculating group cohomology using compactly supported cohomology below.

Observe that for a connected infinite CW complex Y , $H_c^0(Y; \mathbb{Z}) = \ker \delta_c^0$, which is the set of finitely supported 1-cochains that are constant. So it has to be trivial. (It would be $\mathbb{Z}$ if Y is finite.)

Example 3.8 (Compactly supported cohomology of $\mathbb{R}$). Let $\mathbb{R}$ be cellulated as an infinite graph. Let v_i be the vertices and let $e_i = \{v_i, v_{i+1}\}$ be the edges.

Since $\mathbb{R}$ is contractible, $H_c^0(\mathbb{R}; \mathbb{Z}) = 0$. For $n > 2$, since there are no n -cells, $H_c^n(\mathbb{R}; \mathbb{Z}) = 0$. Consider now $H_c^1(\mathbb{R}; \mathbb{Z})$. Since there are no 2-cells, all 1-cochains are 1-cocycles. Therefore $H_c^1(\mathbb{R}; \mathbb{Z}) = C_c^1(\mathbb{R}; \mathbb{Z}) / \text{im } \delta_c^0$.

Let v_i^*, e_i^* be the dual cells of e_i, v_i respectively. Consider the augmented chain $C_c^0(\mathbb{R}; \mathbb{Z}) \xrightarrow{\delta_c^0} C_c^1(\mathbb{R}; \mathbb{Z}) \xrightarrow{\epsilon} \mathbb{Z}$ where ϵ takes every e_i^* to 1. Take $\alpha \in C_c^0(\mathbb{R}; \mathbb{Z})$. We can write α as $\sum_i a_i v_i^*$. Then $\delta_c^0(\alpha) = \sum_i a_i (\delta_c^0(v_i^*))$. But $\delta_c^0(v_i^*) = e_i^* - e_{i+1}^*$, so $\epsilon(\delta_c^0(\alpha)) = \sum_i a_i (1 - 1) = 0$. That is, $\text{im } \delta_c^0 \subseteq \ker \epsilon$. As a result, ϵ factors through $\bar{\epsilon} : C_c^1(\mathbb{R}; \mathbb{Z}) / \text{im } \delta_c^0 = H_c^1(\mathbb{R}; \mathbb{Z}) \rightarrow \mathbb{Z}$.

Clearly $\bar{\epsilon}$ is surjective. To show that $\bar{\epsilon}$ is injective, we need to prove that if $\sum_{i=m}^n a_i = 0$, then $\sum_{i=m}^n a_i e_i^* \in \text{im } \delta_c^0$. We remark that it's important that $\sum_{i=m}^n a_i$ is a finite sum. Notice that

$$\cdots + a_{n-1} e_{n-1}^* + a_n e_n^* + a_n \delta_c^0(v_{n-1}^*) = \cdots + (a_{n-1} + a_n) e_{n-1}^* + 0 \cdot e_n^*,$$

so by adding some coboundary consecutively, we can get a cochain supported on a single edge with coefficient zero. Thus the original cochain is also a coboundary.

Corollary 3.9.

$$H_c^i(\mathbb{Z}; \mathbb{Z}[\mathbb{Z}]) = \begin{cases} \mathbb{Z} & i = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Example 3.10 (Compactly supported cohomology of infinite 4-valence tree). Consider the infinite 4-valence tree T (the universal cover of the wedge of two circles). Similarly, $H_c^i(T; \mathbb{Z}) = 0$ when $i \neq 1$. For $H_c^1(T; \mathbb{Z})$, by Remark 3.7 it is the direct limit of $H^1(T, T \setminus K; \mathbb{Z})$ over all compact sets. Consider the long exact sequence of reduced cohomology groups:

$$\tilde{H}^0(T; \mathbb{Z}) \rightarrow \tilde{H}^0(T \setminus K; \mathbb{Z}) \rightarrow \tilde{H}^1(T, T \setminus K; \mathbb{Z}) \rightarrow \tilde{H}^1(T; \mathbb{Z}).$$

Since T is contractible, $\tilde{H}^i(T; \mathbb{Z}) = 0$, hence $\tilde{H}^0(T \setminus K; \mathbb{Z}) \cong \tilde{H}^1(T, T \setminus K; \mathbb{Z})$. Now $\tilde{H}^0(T \setminus K; \mathbb{Z})$ counts the components of $T \setminus K$, which increases infinitely as the size of K increases.

We conclude that $\tilde{H}^1(T, T \setminus K; \mathbb{Z})$ is infinitely generated. In fact, it's an infinitely generated free abelian group.

Corollary 3.11.

$$H_c^i(\mathbb{F}_2; \mathbb{Z}[\mathbb{F}_2]) = \begin{cases} D & i = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where D is infinitely generated.

Observe that the argument of Example 3.10 works for any contractible space. That is, in general, 1-cohomology of a group with group-ring coefficients has rank of numbers of ends of the group plus 1.

4. GROUP COHOMOLOGY OF SURFACE GROUPS

Example 4.1 (Compactly supported cohomology of R^n). Suppose $\mathbb{R}^n$ is equipped with a polyhedral decomposition. Suppose each $(n-1)$ -cell is adjacent to 2 oriented n -cells, and the $n-1$ -cell appears in their boundary with opposite orientation. Suppose further that each compact $K \subseteq \mathbb{R}^n$ is contained in a ball. (For example, regard R^3 as the universal covering space of 3-torus.)

Since $\mathbb{R}^n$ is contractible, $H_c^0(\mathbb{R}^n; \mathbb{Z}) = 0$.

For $1 \leq i < n$, take $\eta \in C_c^i(\mathbb{R}^n; \mathbb{Z})$ such that $\delta\eta = 0$. Take a ball B such that $\text{supp}(\eta) \subseteq B$. We can regard η as an element of $H^i(\mathbb{R}^n, \mathbb{R}^n \setminus B)$ (that is, a cochain on $\mathbb{R}^n$ that vanishes on B). Recall that $H^i(\mathbb{R}^n, \mathbb{R}^n \setminus B) \cong \tilde{H}^{i-1}(\mathbb{R}^n \setminus B) \cong \tilde{H}^{i-1}(\mathbb{S}^{n-1}) = 0$.

Consider now the augmented chain $C_c^{n-1}(\mathbb{R}^n; \mathbb{Z}) \xrightarrow{\delta} C_c^n(\mathbb{R}^n; \mathbb{Z}) \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$. The assumption that each $(n-1)$ -cell occurs in the boundary of two n -cells with opposite orientation implies that $\text{im } \delta \subseteq \ker \epsilon$ (similar to Example 3.8), so ϵ factors through $H_c^n(\mathbb{R}^n; \mathbb{Z}) \rightarrow \mathbb{Z}$. Again, we need to show that the map is injective. The argument is similar to Example 3.8: we take the dual graph of $\mathbb{R}^n$ so that vertices are dual n -cells and edges are dual $(n-1)$ -cells. We then adjust one dual n -cell at a time until the cochain is supported on a single dual n -cell with coefficient zero.

Corollary 4.2. *Let M be a closed n -manifold with $\widetilde{M} \cong \mathbb{R}^n$. Let $G = \pi_1(M)$. Then*

$$H^i(G; \mathbb{Z}G) \cong H_c^i(\mathbb{R}^n; \mathbb{Z}) = \begin{cases} \mathbb{Z} & i = n, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, if Σ_g is a closed orientable surface with genus $g \geq 1$ and $G = \pi_1(\Sigma_g)$, then

$$H^i(G; \mathbb{Z}G) = \begin{cases} \mathbb{Z} & i = 2, \\ 0 & \text{otherwise.} \end{cases}$$

Remark 4.3. For non-orientable manifolds, the cohomology would be the same. What's different is the action of Γ on the top cohomology $\mathbb{Z}$.

In general, $H^i(G; \mathbb{Z}G)$ is a $\mathbb{Z}G$ -module. Take a free resolution F . A priori, $\text{Hom}_{\mathbb{Z}G}(F_i, \mathbb{Z}G)$ is just an abelian group, but the right action by G satisfies the criterion from Lemma 3.3. Therefore, as a quotient of $\mathbb{Z}G$ -module by $\mathbb{Z}G$ -module, $H^i(G; \mathbb{Z}G)$ is also a $\mathbb{Z}G$ -module.

Question 4.4 (Wall). Suppose G has type F (recall Definition 1.11) and suppose that for some n , $H^n(G; \mathbb{Z}G) = \mathbb{Z}$ and $H^i(G; \mathbb{Z}G) = 0$ otherwise. Does there exist a closed aspherical manifold M such that $G = \pi_1(M)$?

Question 4.4 is known to be true for $n = 1$ ($G = \mathbb{Z}$) and $n = 2$ (surface groups). It is wide open for $n \geq 3$.

Notice that if such an M exists, it's unique up to homotopic equivalence by the uniqueness of $K(G, 1)$. It's natural to ask:

Question 4.5 (Borel conjecture). If M_1, M_2 are closed aspherical manifolds that are homotopic equivalent to each other, are they homeomorphic to each other?

Question 4.5 is known to be true for $n = 1, 2, 3$. Note that in general, without the assumption of aspherical, this is not true.

Recall that two homotopic equivalent manifolds M_1, M_2 have isomorphic fundamental groups. A group theoretic version of Question 4.5 is: Given a group G , does it have a unique $K(G, 1)$ up to homeomorphism?

Theorem 4.6. *Let X be a CW complex such that $\tilde{X}$ is k -connected (recall Definition 2.1) for some k . Then there exists an aspherical CW complex Z such that $Z^{k+1} = X^{k+1}$.*

Theorem 4.6 is the reason we are interested in 1-connected spaces, because we can obtain a CW complex with the same fundamental group.

Proof. Assume inductively that we've built Z_n such that $\tilde{Z}_n$ is n -connected and $Z_n^{k+1} = X^{k+1}$ (Z_k is just X). Note that every map $f : \mathbb{S}^{n+1} \rightarrow Z_n$ is homotopic to a map with image in Z_n^{n+1} . Let $\{F_\alpha : \mathbb{S}^{n+1} \rightarrow Z_n^{n+1} \mid \alpha \in A_n\}$ denote the homotopy classes of maps $\mathbb{S}^{n+1} \rightarrow Z_n$. Build Z_{n+1} by gluing $(n+1)$ -cells to Z_n using F_α s as attaching maps.

Observe that $Z_{n+1}^{n+1} = Z_n^{n+1}$. Any $g : \mathbb{S}^{n+1} \rightarrow Z_{n+1}$ is homotopic to some F_α , hence is null-homotopic. Therefore $\tilde{Z}_{n+1}$ is $(n+1)$ -connected. $\square$

5. PROJECTIVE MODULES AND MORE FINITENESS PROPERTIES

Recall Theorem 3.6. In fact, to obtain the isomorphism $H^i(G; \mathbb{Z}G) \cong H_c^i(\tilde{X}; \mathbb{Z})$, we need only finite X^{i+1} .

Definition 5.1. Let G be a group. The *cohomological dimension* of G , denoted as $\text{cd}(G)$ (or $\text{cd}_{\mathbb{Z}}(G)$), is defined to be

$$\sup\{n \mid \exists M \text{ a } \mathbb{Z}G\text{-module such that } H^n(G; M) \neq 0\}.$$

If there's no bound on n , then we say $\text{cd}(G) = \infty$.

Recall geometric dimension of a group (See Definition 1.11). Given a $K(G, 1)$ of dimension n , we obtain a free resolution of $\mathbb{Z}$ by $\mathbb{Z}G$ -modules that stops in degree n . Consider its cohomology: there is no cochain groups in dimension above n , so the cohomology groups in dimension above n are trivial. That is, the geometric dimension is an upper bound of the cohomological dimension.

Question 5.2. Are there any G with $\text{cd}(G) < \text{gd}(G)$?

Theorem 5.3 ([EG57]). *Let G be a group. Let $k = \max\{\text{cd}(G), 3\}$. Then there exists a $K(G, 1)$ of dimension k .*

So if $\text{cd}(G) \geq 3$, then $\text{cd}(G) = \text{gd}(G)$.

In lower dimensions, $\text{cd}(G) = 0$ if and only if G is trivial if and only if $\text{gd}(G) = 0$. (This is a good exercise). At dimension 1, we have $\text{cd}(G) = 1$ if and only if G is free if and only if $\text{gd}(G) = 1$. However, the proof is significantly harder.

Theorem 5.4 (Stallings-Swan theorem). *If $\text{cd}(G) = 1$ then G is free.*

The case at 2 is still an open question, though there is already a potential counterexample. TODO add reference to the potential counterexample.

Definition 5.5. A $\mathbb{Z}G$ -module P is *projective* if for every exact

$$M_0 \xrightarrow{i} M_1 \xrightarrow{j} M_2,$$

and every map $\phi : P \rightarrow M_1$ with $j\phi = 0$, there exists $\psi : P \rightarrow M_0$ such that $i\psi = \phi$.

$$\begin{array}{ccccc} & & P & & \\ & \swarrow \psi & \downarrow \phi & \searrow 0 & \\ M_0 & \xrightarrow{i} & M_1 & \xrightarrow{j} & M_2 \end{array}$$

Lemma 5.6. *Free modules are projective.*

Proof. Trace the basis of the free module. □

Theorem 5.7. *Projective resolutions are unique up to chain homology.*

Lemma 5.8. *Let P be a projective module and let Q be a direct summand of P . Then Q is projective.*

Proof. Suppose $P = Q \oplus M$.

$$\begin{array}{ccccc} & & P = Q \oplus M & & \\ & \swarrow \psi_P & \uparrow \iota \downarrow \pi & \searrow 0 & \\ & & Q & & \\ & \swarrow \psi_P \iota & \downarrow \phi & \searrow 0 & \\ M_0 & \xrightarrow{i} & M_1 & \xrightarrow{j} & M_2 \end{array}$$

□

Corollary 5.9. *Direct summands of free modules are projective.*

Proposition 5.10. *TFAE:*

- (1) P is projective.
- (2) For every surjective $\pi : M \rightarrow \overline{M}$ and every $\phi : P \rightarrow \overline{M}$, there exists $\psi : P \rightarrow M$ such that $\pi\psi = \phi$.

- (3) *Every exact $0 \rightarrow M \rightarrow M' \rightarrow P \rightarrow 0$ splits.*
 (4) *P is a direct summand of a free module.*

Proof. (1) $\implies$ (2): Replace M by $\ker j$ in Definition 5.5.

(2) $\implies$ (3): Lift $\text{Id} : P \rightarrow P$ to $P \rightarrow M'$.

(3) $\implies$ (4): Exhibits P as a quotient of a free module, then the following SES splits:

$$0 \rightarrow \ker \epsilon \rightarrow F \xrightarrow{\epsilon} P \rightarrow 0.$$

(4) $\implies$ (1): Corollary 5.9. □

Definition 5.11. Let G be a group.

We say G has type FF (resp. FP) if there exists a finite resolution of $\mathbb{Z}$ by finitely generated free (resp. projective) $\mathbb{Z}G$ -modules.

We say G has type FF_n (resp. FP_n) if there exists a resolution

$$\cdots \rightarrow M_1 \rightarrow M_0 \rightarrow \mathbb{Z} \rightarrow 0$$

of $\mathbb{Z}$ by free (resp. projective) $\mathbb{Z}G$ -modules with finitely-generated M_i for $i \leq n$.

We say G has type FF_∞ (resp. FP_∞) if G has type FF_n (resp. FP_n) for all n .

Theorem 5.12. *Type FF implies type FP .*

Question 5.13 (Serre's Conjecture). Does type FP imply type FF ?

Recall type F, F_n, F_∞ (see Definition 1.11). Clearly $F_n \implies FF_n \implies FP_n$. Recall also Theorem 1.9.

Theorem 5.14. *There exists G of type FP_2 which is not finitely presented (hence is not of type F_2)*

Right-angled Artin groups is a class of groups that can tells those properties apart.

Definition 5.15 (Right-angled Artin group). Let Γ be a simplicial graph with vertex set V and edge set E . We associate a group A_Γ to Γ

$$A_\Gamma := \langle V \mid vw = wv \text{ if } \{v, w\} \in E \rangle.$$

Example 5.16. (1) If $|V| = m$ and $E = \emptyset$, then $A_\Gamma = \mathbb{F}_m$.

(2) If $|V| = m$ and Γ is a complete graph, then $A_\Gamma = \mathbb{Z}^m$.

(3) Say $V = \{a, b, c\}$ and $E = \{\{a, b\}\}$, then $A_\Gamma = \mathbb{Z}^2 * \mathbb{Z}$.

Theorem 5.17. *Let k be the clique number (i.e. maximal size of complete subgraph) of Γ . Then $\text{gd}(A_\Gamma) = k$.*

Remark 5.18. Notice that under the assumption of Theorem 5.17, $\mathbb{Z}^k \subseteq A_\Gamma$, so $\text{cd}(A_\Gamma) \geq k$. We are interested in the subgroups of A_Γ and we will investigate the map $\phi_\Gamma : A_\Gamma \rightarrow \mathbb{Z}$, defined by $v \mapsto 1, \forall v \in V$.

6. THE BESTVINA-BRADY THEOREM FOR FLAG COMPLEXES

Definition 6.1 (flag simplicial complex). A simplicial complex Γ is *flag* if whenever a set of vertices $\{v_0, \dots, v_k\}$ span a complete subgraph, then they span a k -simplex.

In other words, every non-simplex contains a non-edge.

Any graph Γ determines a flag simplicial complex.

Definition 6.2 (RAAGs of flag simplicial complex). Let Γ be a simplicial graph with vertex set V and edge set E . Let A_Γ be the associated RAAG (See Definition 5.15). If L is a flag simplicial complex, Then $A_L := A_{L(1)}$.

Define $\phi_L : A_L \rightarrow \mathbb{Z}$ by $v \mapsto 1$. Let $K_L = \ker \phi_L$.

Remark 6.3. Since ϕ_L is a homomorphism, K_L is related to the “topology” of flag of Γ .

Example 6.4. Let L be a square, then $A_L = \mathbb{F}_2 \times \mathbb{F}_2$.

In general, join of graphs corresponds to product of groups. (Disjoint union corresponds to free product).

Definition 6.5. A space X is *acyclic* if $\tilde{H}_i(X) = 0$ for all $i \geq 0$.

Recall that a group is *perfect* if it has trivial abelianization, i.e. $\tilde{H}_1(X)$ is trivial. By definition, every acyclic group is perfect.

Definition 6.6. A space X is *homologically (k) -connected* if $\tilde{H}_i(X) = 0$ for $i \leq k$

Theorem 6.7 (The Bestvina-Brady Theorem). *Let L be a finite flag simplicial complex.*

- (1) K_L is finitely presented if and only if L is simply connected.
- (2) K_L has type FP if and only if L is acyclic.
- (3) K_L has type FP_n if and only if L is homologically $(n-1)$ -connected.

Theorem 6.8 ([Hat01, Theorem 4.32]). *If X is $(n-1)$ -connected, $n \geq 2$, then $\tilde{H}_i(X) = 0$ for $0 \leq i \leq n-1$, and $H_n(X) \cong \pi_n(X)$.*

Theorem 6.8 will guide our search for a non-contractible acyclic space. We also want π_1 to be perfect.

Example 6.9 (Poincare’s homology sphere). Take an empty dodecahedron such that opposite sides are the same up to a twist. Identify opposite sides with a $\pi/5$ twist, then we get the 2-skeleton of Poincare’s homology sphere, denoted as X .

Let M be the homology sphere (which is a 3-manifold, and has X as its 2-skeleton). We have $\tilde{H}_3(M) \cong \tilde{H}_3(\mathbb{S}^3) = \mathbb{Z}$, and that $\tilde{H}_i(M) \cong \tilde{H}_i(\mathbb{S}^3) = 0$ for all other i ’s.

Recall Universal coefficient: if $H_i(M)$ is torsion free, then $H^{i+1}(M; \mathbb{Z}) \cong \text{Hom}(H_i(M), \mathbb{Z})$.

Recall Poincare Duality: $H_i(M) \cong H^{3-i}(M; \mathbb{Z})$

Since $H_0(M) = 0$, $H_1(M) = 0 \implies H^1(M; \mathbb{Z}) = 0$ by Universal Coefficient theorem, then $H_2(M) = 0$ (by Poincare Duality).

So do Cellular chain complex for M we get

$$0 \rightarrow C_3(M) \xrightarrow{\alpha_3} C_2(M) \rightarrow C_1(M) \rightarrow \mathbb{Z} \rightarrow 0$$

Observe that $\alpha_3 = 0$ because M has no boundary after gluing. Also $C_3(M) = \mathbb{Z}$ because there's only one 3-cell.

We can ignore $C_3(M)$. The rest of the chain (and the homology) coincides with that of X :

$$0 \rightarrow C_2(X) \rightarrow C_1(X) \rightarrow \mathbb{Z} \rightarrow 0$$

And so all homologies of X are trivial.

This X with K_X gives the first example of a group which is acyclic but non-contractible complex.

Now subdivide the faces to (10) triangles to obtain a flag complex L . This L is not simply connected, so K_L is not finitely presented. It is still acyclic, so K_L has type FP . This gives the first example of a group that has FP but is not finitely presented.

Remark 6.10. If we do $2\pi/5$ rotation with filled dodecahedron, then we get an infinite perfect π_1 , and it's hyperbolic.

One can see that by seeing how there are 5 pentagons around an edge, so angle argument gives the hyperbolic geometry.

Our goal now is to prove Theorem 6.7. We first introduce yet another finiteness property.

Definition 6.11. A group G has type FH_n (resp. FH) if there exists a homologically $(n-1)$ -connected (resp. acyclic) complex X with a free cellular cocompact G action.

We now have four types of finiteness properties: F, FP, FF, FH (See Definitions 1.11, 5.11 and 6.11).

We know that $F_n \implies FH_n \implies FF_n \implies FP_n$. We know there exists a group that has type FP_2 but not type F_2 (Recall Theorem 5.14). Lemma 6.12 proves $FP_n \implies FF_n$.

Similarly, $F \implies FH \implies FF \implies FP$ (The last part is Theorem 5.12). It's still an open question whether $FP \implies FF$ (See Question 5.13).

Lemma 6.12. $FP_n \implies FF_n$.

Proof. Let M be a projective $\mathbb{Z}G$ -module. Take M as a quotient of free module:

$$0 \rightarrow K \rightarrow F \rightarrow M \rightarrow 0.$$

If M is finitely generated, we can assume F is also finitely generated. By Proposition 5.10 K is projective and the sequence splits, which forces K to be finitely generated. Hence $F = K \oplus M$.

Take a finitely generated projective resolution $\{P_i\}_{i=0}^n$ of $\mathbb{Z}$. Then there exists $F_0 = P_0 \oplus Q_0$. Build:

$$P_1 \oplus Q_0 \rightarrow P_0 \oplus Q_0 \rightarrow \mathbb{Z}$$

As a direct sum of projective modules, $P_1 \oplus Q_0$ is also projective. We can then find Q_1 such that $Q_1 \oplus P_1 \oplus Q_0$ is a finitely generated free module. We can repeat this procedure to eventually obtain a finitely generated free resolution. $\square$

We make some observations below.

- (1) If there exists $n-1$ -connected X with a free cocompact G action, then G has type F_n by turning $G \backslash X$ into a $K(G, 1)$. This proves $F_n \implies FH_n$.
- (2) $FH_n \implies FF_n$ Take the X you get, $C_i(X)$ are finitely generated-free $\mathbb{Z}G$ -modules, and this is exact because it's homologically n -connected.
- (3) $FH_1 \implies F_1$ because 0th homology is about being connected.
- (4) $FF_1 \implies F_1$ See Lemma 7.4.

In particular, we have:

$$F_1 \iff FF_1 \iff FH_1 \iff \text{finitely generated.}$$

7. SCHANUEL'S LEMMA AND THE SALVETTI COMPLEX

Lemma 7.1 (Schanuel's Lemma). *Suppose P, P' are projective modules, and we have exact sequences*

$$0 \rightarrow K \rightarrow P \xrightarrow{\pi} M \rightarrow 0,$$

$$0 \rightarrow K' \rightarrow P' \xrightarrow{\pi'} M \rightarrow 0.$$

Then $P \oplus K' \cong P' \oplus K$.

Proof. Let $Q = \{(p, p') \in P \oplus P' \mid \pi(p) = \pi'(p')\}$ be the fiber product of π, π' . Define $\alpha : Q \rightarrow P$ by $(p, p') \mapsto p$. Then $\ker \alpha = \{(p, p') \mid p = 0, \pi'(p') = \pi(p) = 0\}$. But that's just $0 \oplus \ker \pi' = K'$. Since π' is surjective, so $\forall p \in P, \exists p' \in P'$ such that $\pi'(p') = \pi(p)$, that is, α is surjective. So the following short sequence is exact:

$$0 \rightarrow K' \rightarrow Q \xrightarrow{\alpha} P \rightarrow 0.$$

Since P is projective, $Q \cong P \oplus K'$ (See Proposition 5.10). By the same argument, $P' \oplus K \cong Q$. $\square$

Corollary 7.2. *Suppose $P_1 \rightarrow P_0 \xrightarrow{\pi} M \rightarrow 0$ is an exact complex where P_0, P_1 are finitely generated projective modules. Suppose also $P'_0 \xrightarrow{\pi'} M \rightarrow 0$ is an exact complex where P'_0 is a finitely generated projective module. Then $\ker \pi'$ is finitely generated.*

Proof. The map π gives a short exact sequence:

$$0 \rightarrow \ker \pi \rightarrow P_0 \xrightarrow{\pi} M \rightarrow 0.$$

Notice that as a submodule of a finitely generated module, $\ker \pi$ is also finitely generated.

We also have:

$$0 \rightarrow \ker \pi' \rightarrow P'_0 \xrightarrow{\pi'} M \rightarrow 0.$$

By Lemma 7.1, $\ker \pi \oplus P'_0 \cong \ker \pi' \oplus P_0$. Since $\ker \pi$, P'_0 , P_0 are all finitely generated, so is $\ker \pi'$. $\square$

Recall that groups with type FP_n give rise to finitely generated projective resolutions (See Definition 5.11). Proposition 7.3 is a higher dimension version of ??.

Proposition 7.3. *Let M be a module. Suppose $\{P_i\}_{i=0}^n$ is a finitely generated projective resolution of M . Suppose $\{P'_i\}_{i=0}^k$ is another finitely generated projective resolution and $k < n$. Then $\ker(P'_k \rightarrow P'_{k-1})$ is also finitely generated.*

Recall Definition 5.11 and Definition 1.11.

Lemma 7.4. *Suppose a group G has type FF_1 , then G is finitely generated.*

Proof. If a group G has type FF_1 then there exists a free resolution of $\mathbb{Z}$ with $\mathbb{Z}G$ -module:

$$F_1 \rightarrow F_0 \xrightarrow{\alpha} \mathbb{Z} \rightarrow 0.$$

We also have the map

$$\mathbb{Z}G \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0, \quad g \mapsto 1.$$

By Corollary 7.2, $\ker \epsilon$ is finitely generated as a $\mathbb{Z}G$ -module.

Claim 1. Group G is finitely generated if and only if $\ker \epsilon$ is finitely generated as a $\mathbb{Z}G$ -module.

Let S be a generating set of G , then $\{1 - s \mid s \in S\}$ is a generating set of $\ker \epsilon$. So $\ker \epsilon$ is finitely generated if S is finite.

Suppose now $\ker \epsilon$ is finitely generated. Notice two facts:

- (1) $\ker \epsilon$ is generated by $\{1 - g \mid g \in G\}$, and
- (2) $ab - 1 = a(b - 1) + (a - 1)$.

Then there exists a finite subset $\{1 - s \mid s \in S\}$ of $\{1 - g \mid g \in G\}$ that generates $\ker \epsilon$. Take the Cayley graph Γ of G with respect to S , recall that Γ is connected if and only if S generates G . Now consider the chain complex

$$C_1(\Gamma) \xrightarrow{\partial_1} C_0(\Gamma) \rightarrow \mathbb{Z} \rightarrow 0.$$

Here, $C_0(\Gamma) = \mathbb{Z}G$. We verify that $\partial_1(e) = g - gs = g(1 - s) = 0$. We can show $\ker \epsilon \subseteq \text{im } \partial_1$ using the same strategy as in Example 3.8.

It follows that $\tilde{H}_0(\Gamma) = 0$, i.e. Γ is connected, i.e. G is finitely generated. $\square$

Proposition 7.5. *If G has type FF_2 , then G has type FH_2 . (See Definitions 5.11 and 6.11.)*

Proof. Suppose G has type FF_2 , then G has type FF_1 and is finitely generated by Lemma 7.4. Suppose $G = \langle S | R \rangle$ where S is finite. If R is finite, then we are done. If not, we build a presentation 2-complex Z and let $\tilde{Z}$ be the universal cover. We then have:

$$C_2(\tilde{Z}) \xrightarrow{\partial_2} C_1(\tilde{Z}) \xrightarrow{\partial_1} C_0(\tilde{Z}) \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0,$$

where $C_1(\tilde{Z}), C_0(\tilde{Z})$ are finitely generated free $\mathbb{Z}G$ -module (generated by orbits of i -cells in Z). By an upgraded version of Lemma 7.1, $\ker \partial_1$ is a finitely generated $\mathbb{Z}G$ -module. So there exists a finite subset of 2-cells in Z such that if we add only their orbits to $\tilde{Z}^{(1)}$, we get a space X_0 with finitely many orbits of 2-cells. So X_0 is homologically 1-connected. (Note that X_0 is probably not simply-connected). $\square$

Question 7.6 (Bestvina-Brady). Does FF_n imply FH_n for $n > 2$?

Definition 7.7. Let L be a finite flag simplicial complex and let A_L be the RAAG associated to L . The Salvetti complex is a presentation 2-complex X_L built in the following way:

- X_L has one vertex x_0 ,
- For each $v \in V(L)$, associate an edge in X_L
- For each $\{v, w\} \in E(L)$, associate a square in X_L , whose boundaries correspond to v, w .
- For each simplex in L spanned by $v_0, \dots, v_k$, add a $(k+1)$ -torus.

If L is flag, then $\pi_1(X_L) \cong A_L$ and X_L is aspherical.

FIX package cref breaks and some hyperlinks are wrong. Need to find an alternative.

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